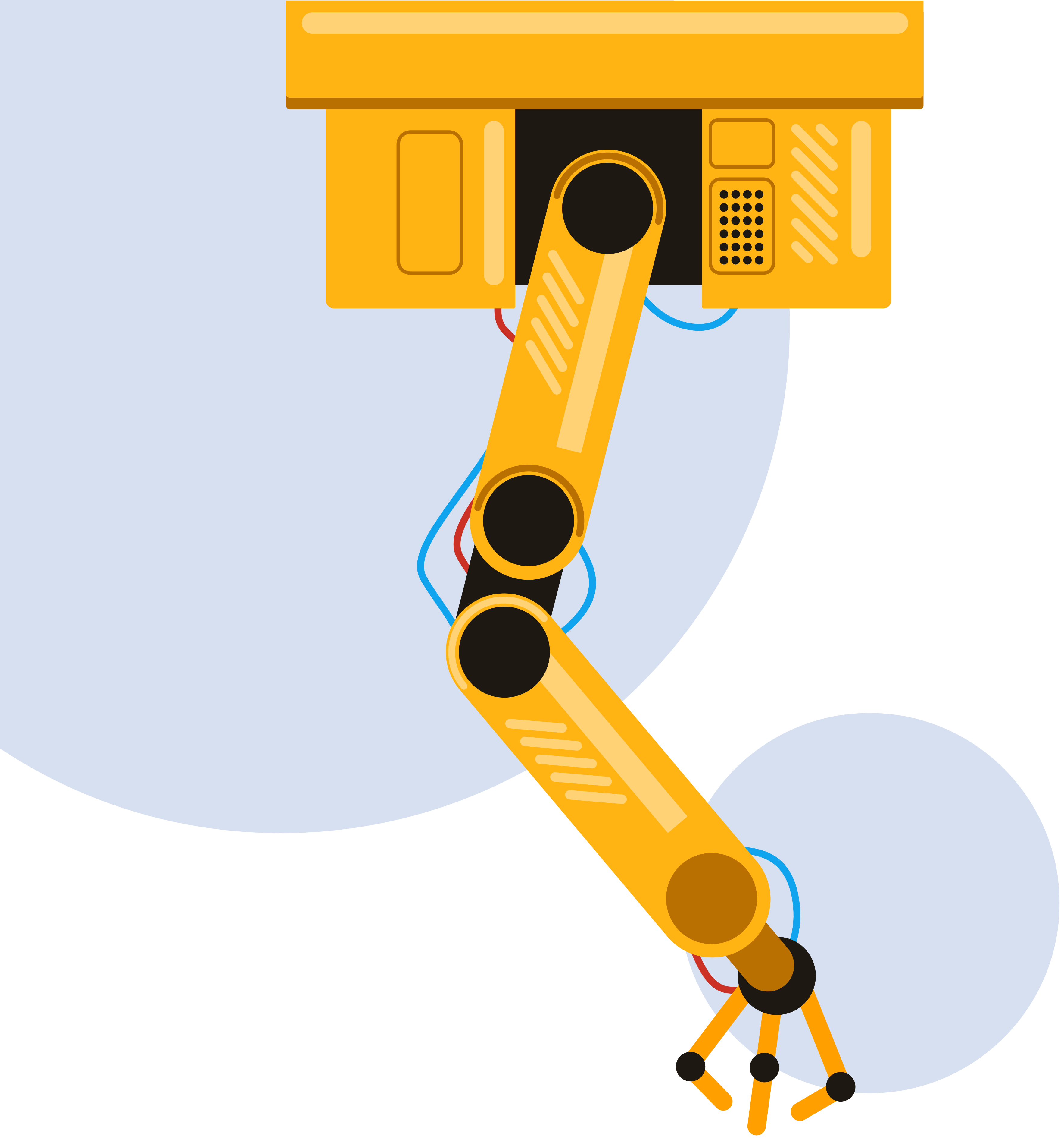
CLASE 10

Robótica



Inteligencia artificial (Teoría)

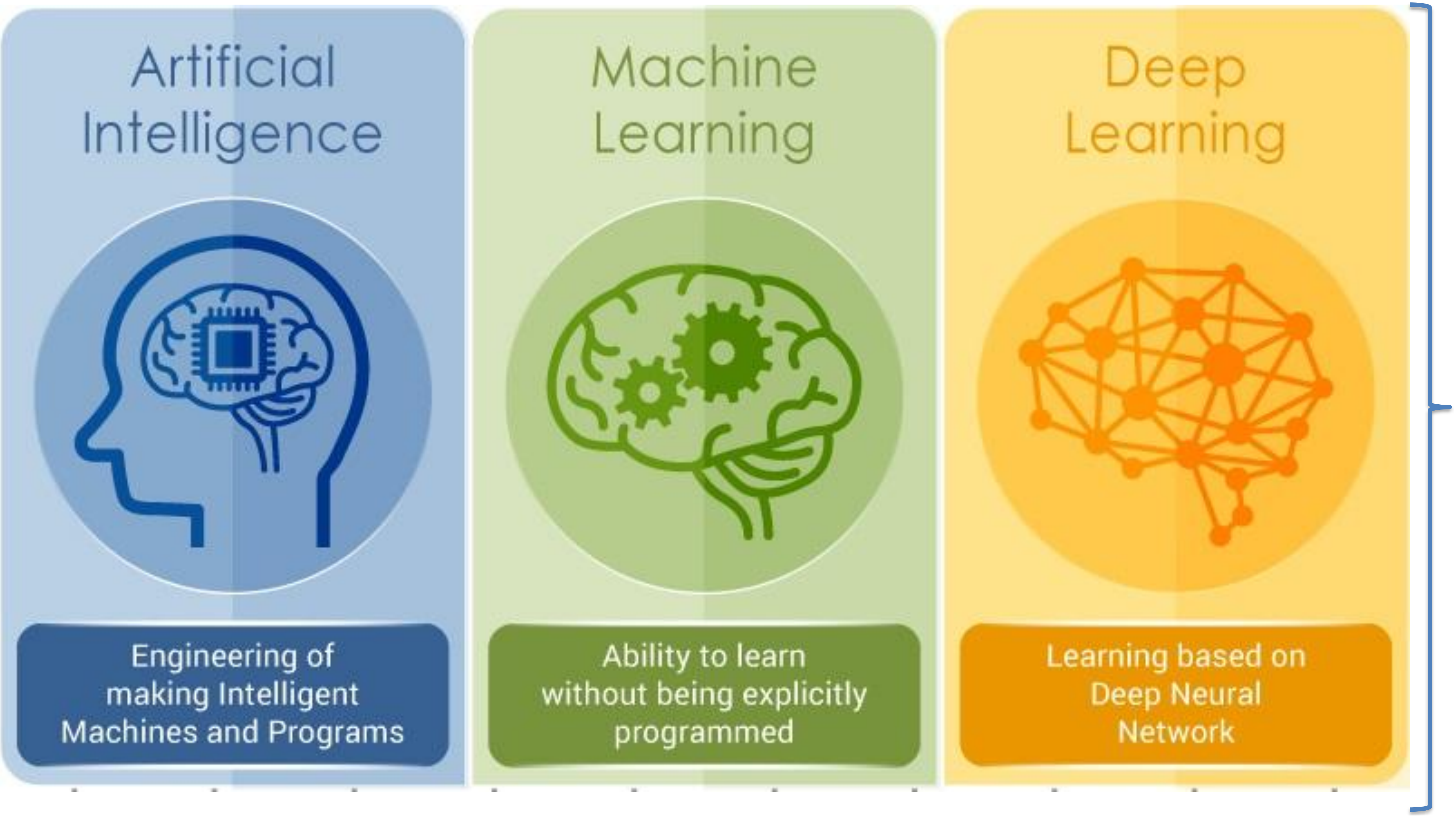
A.1 Historia e introducción

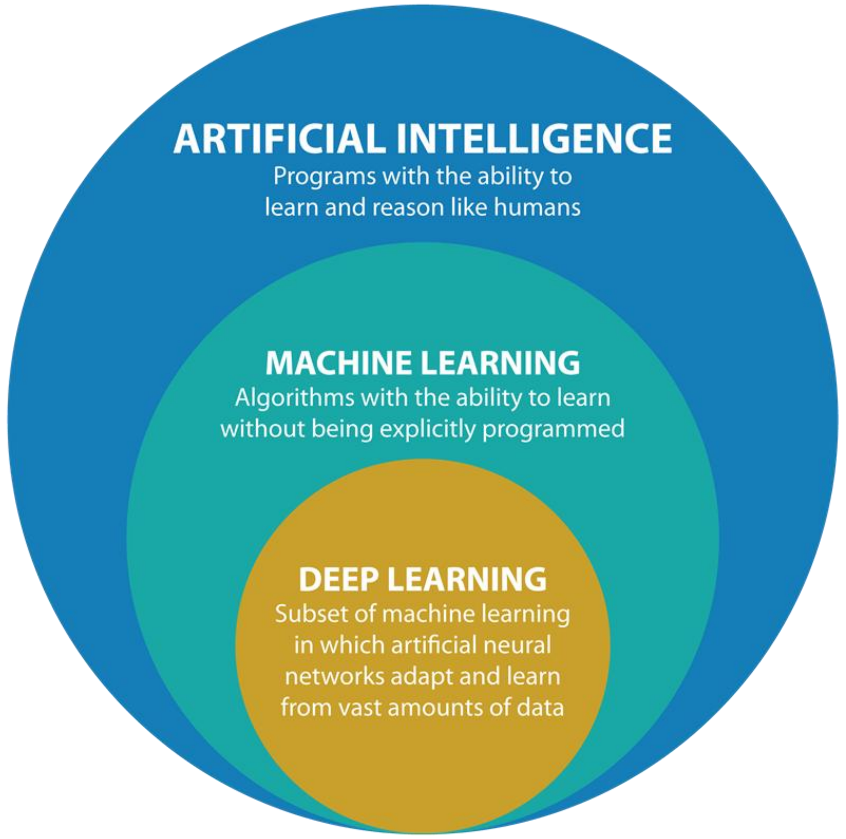
A.2 Clasificación de redes

A.3 Red neuronal convolucional

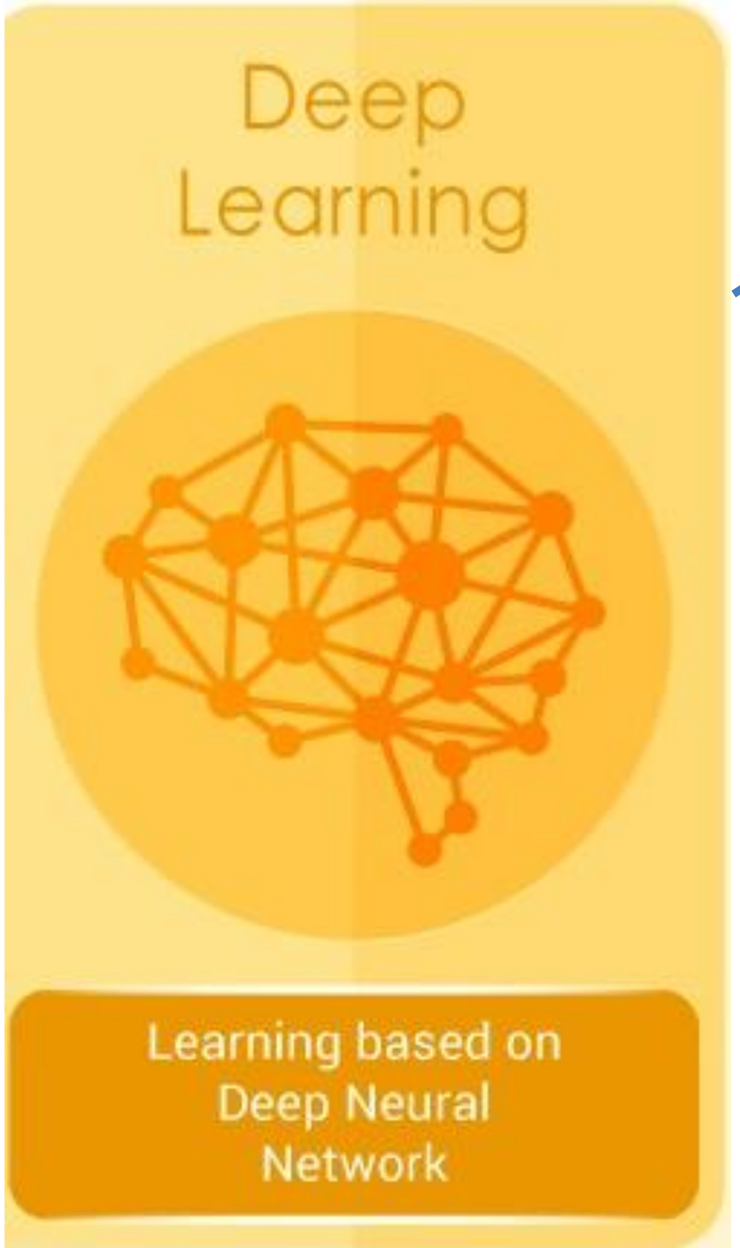
A.4 Métodos de evaluación

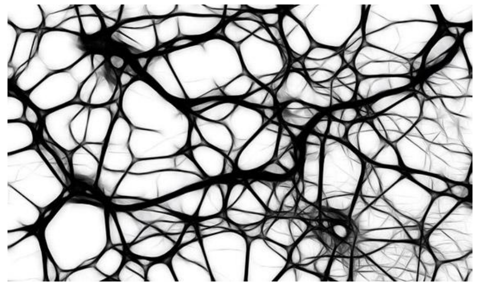
Historia e introducción



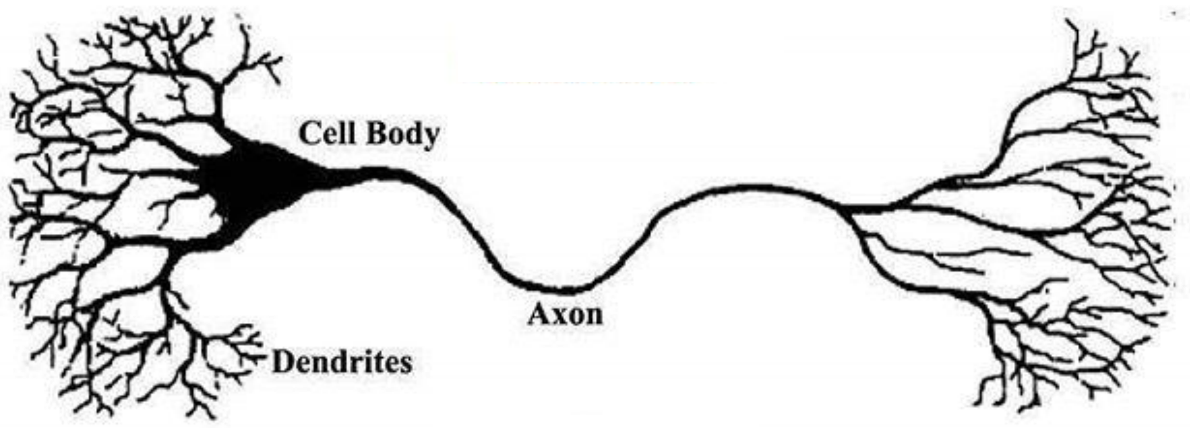


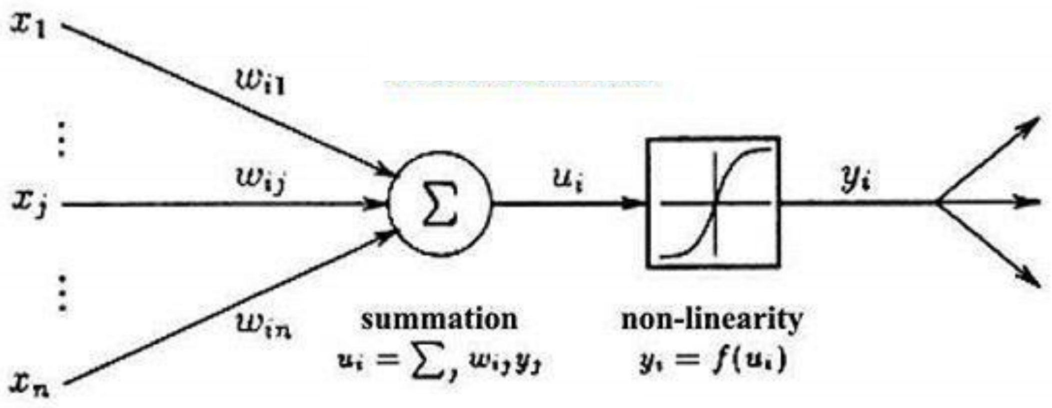
Historia e introducción

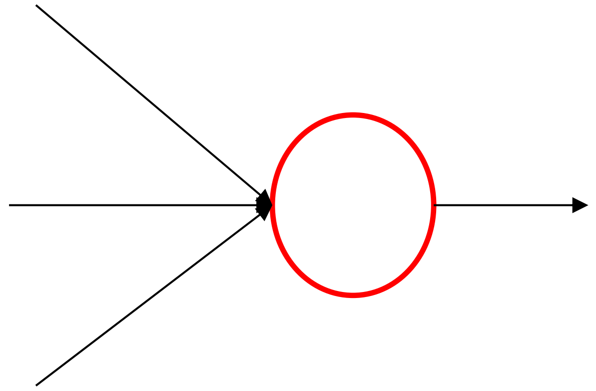




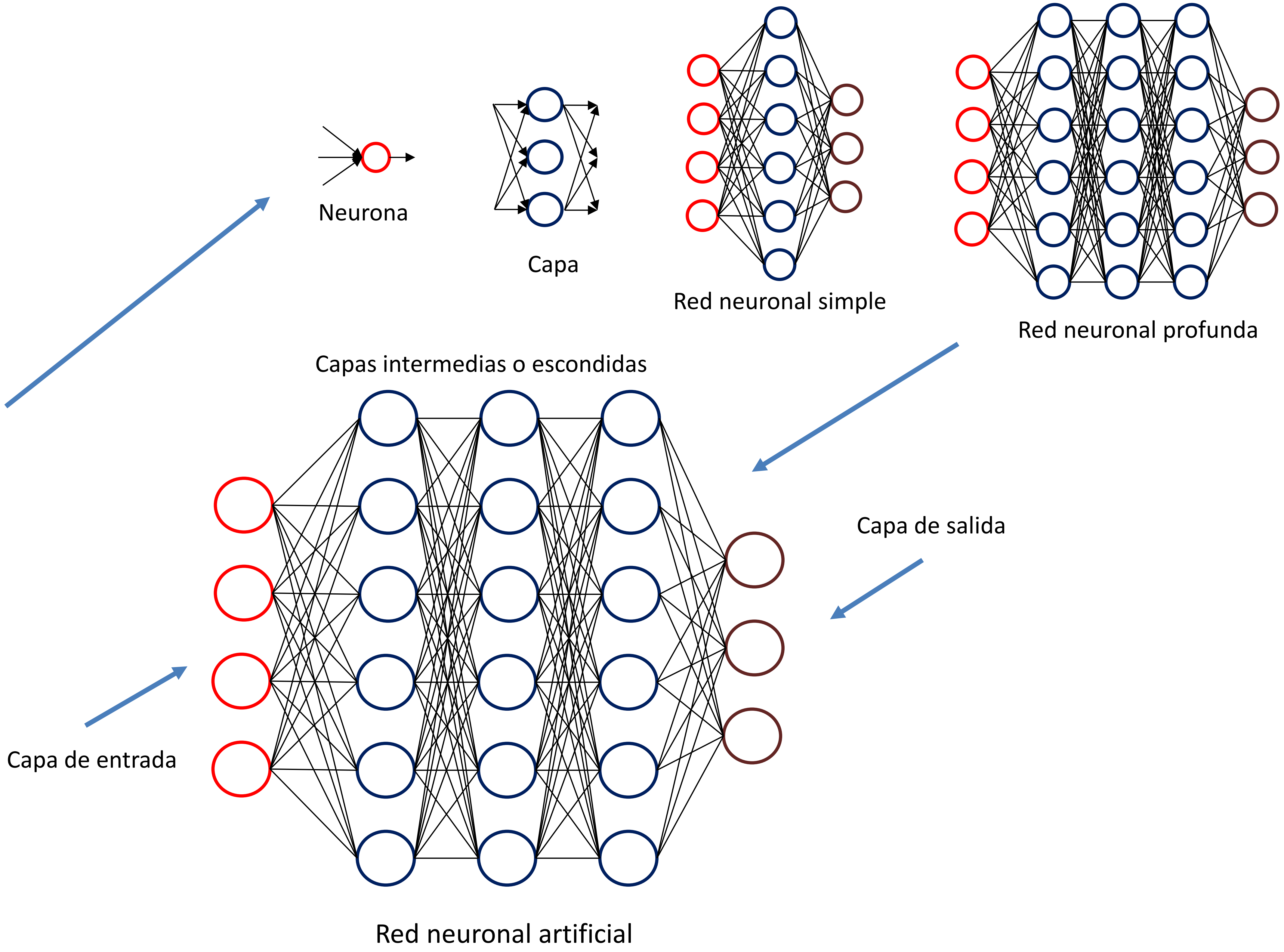
Neuronas Biológicas



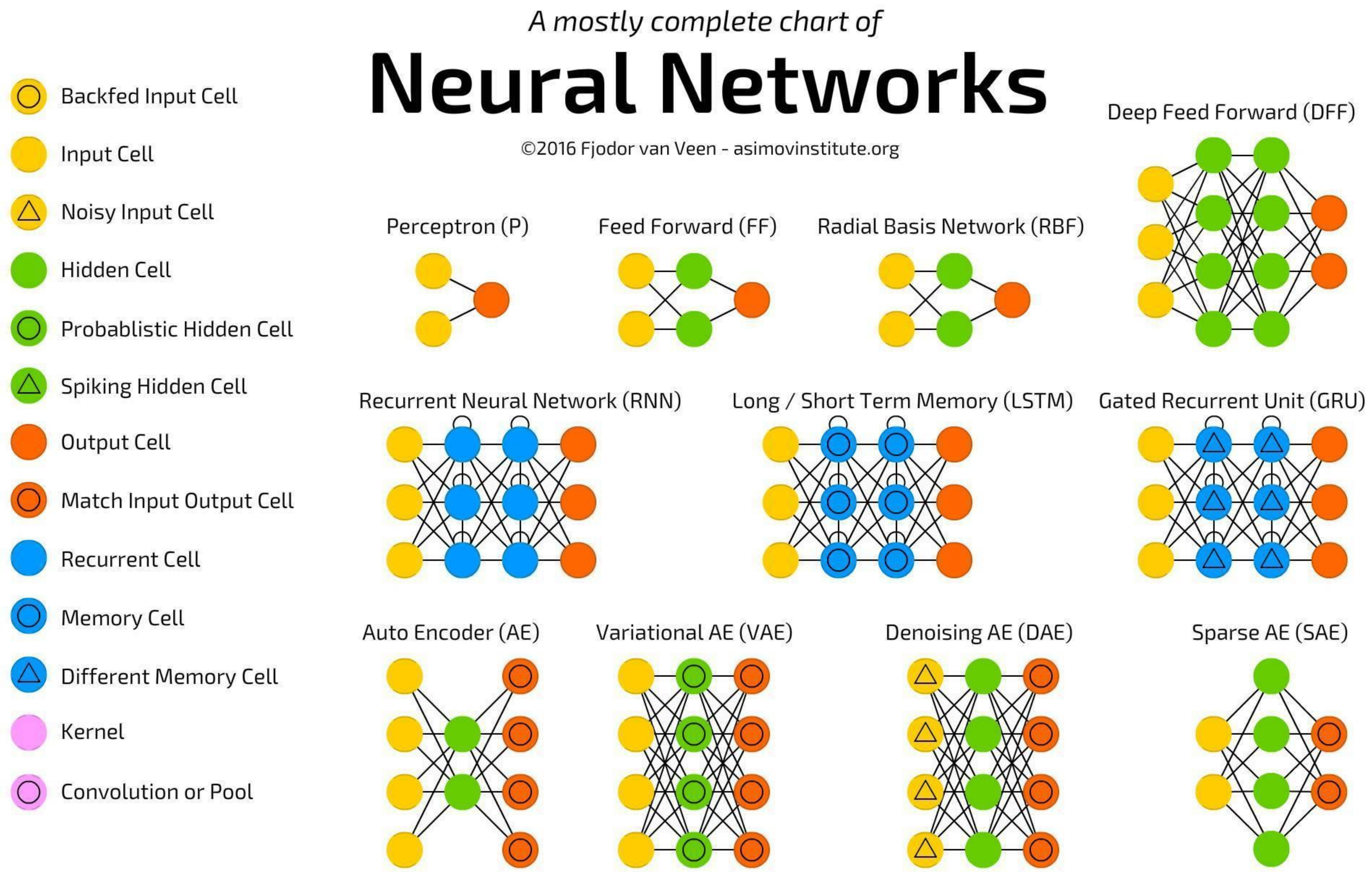




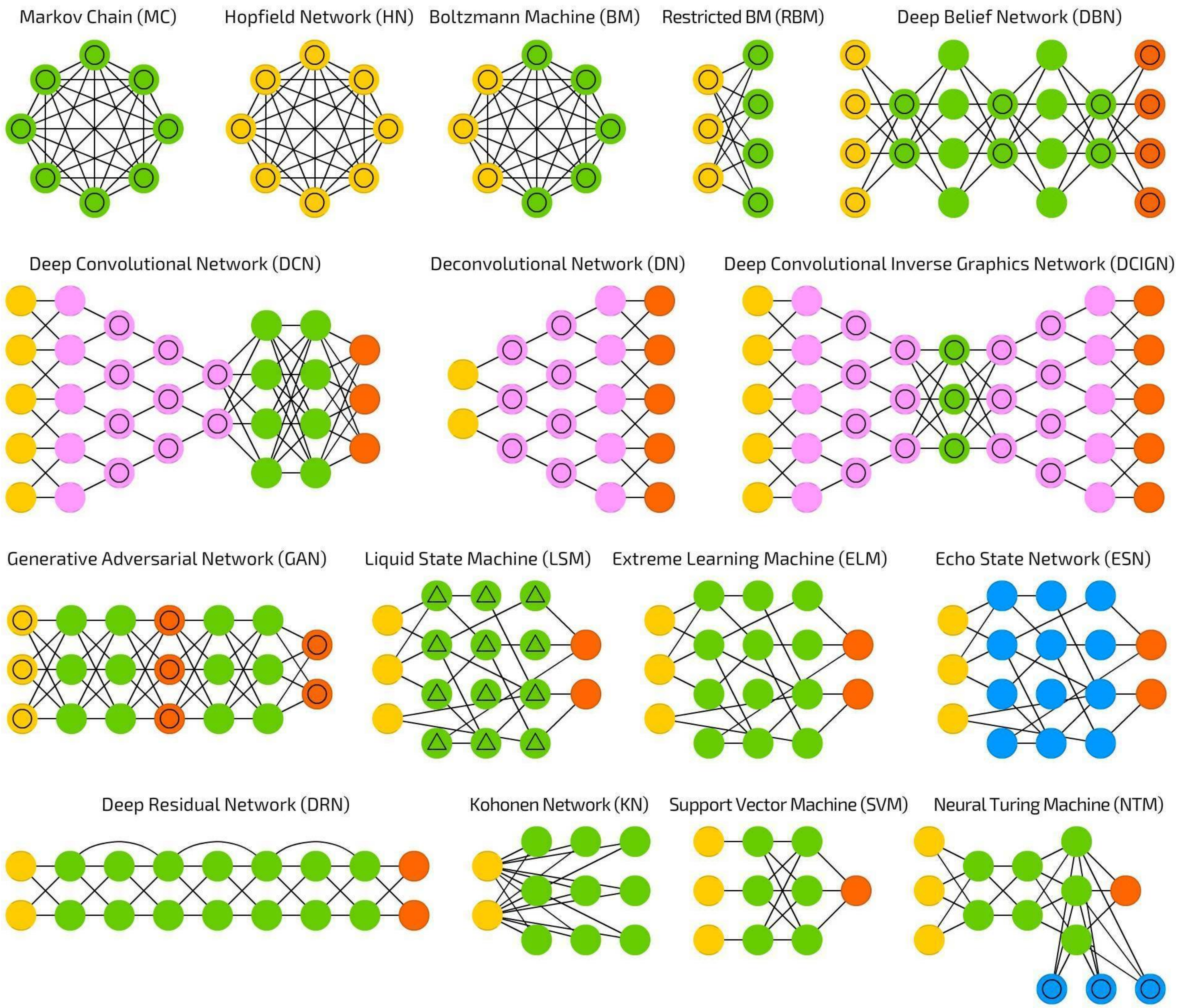
Historia e introducción



Clasificación

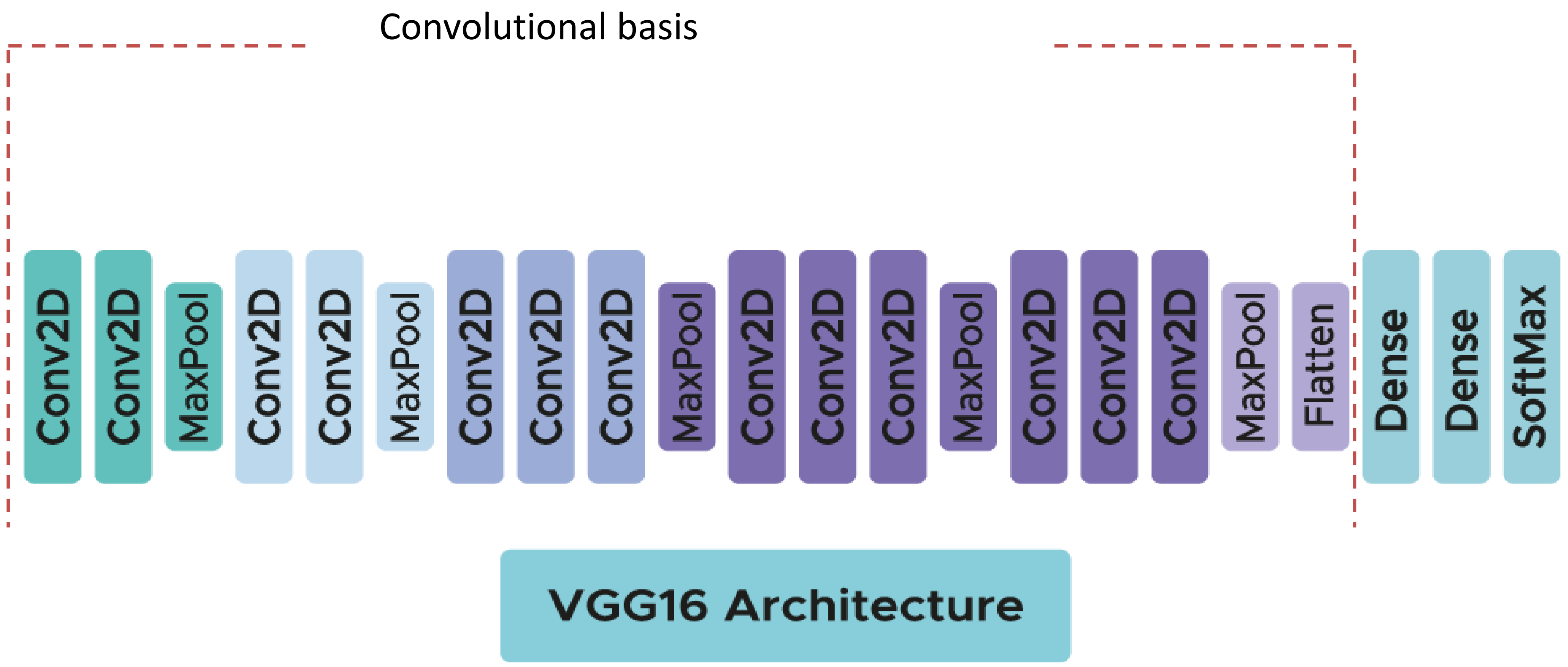


Clasificación

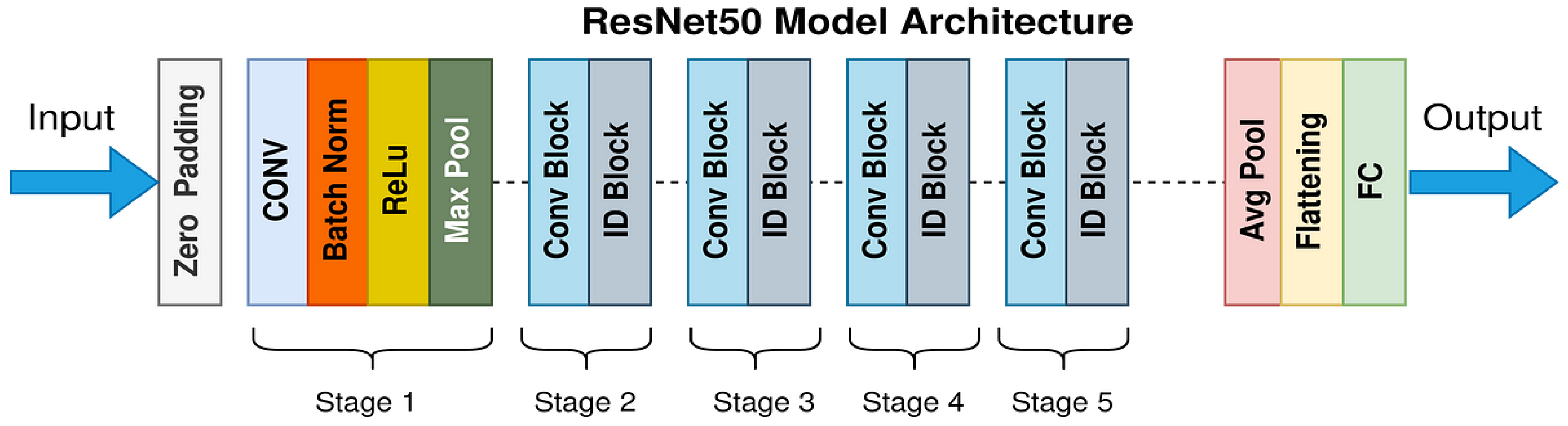


Red neuronal convolucional
CNN

Convolutional blocks of a pre-trained CNN

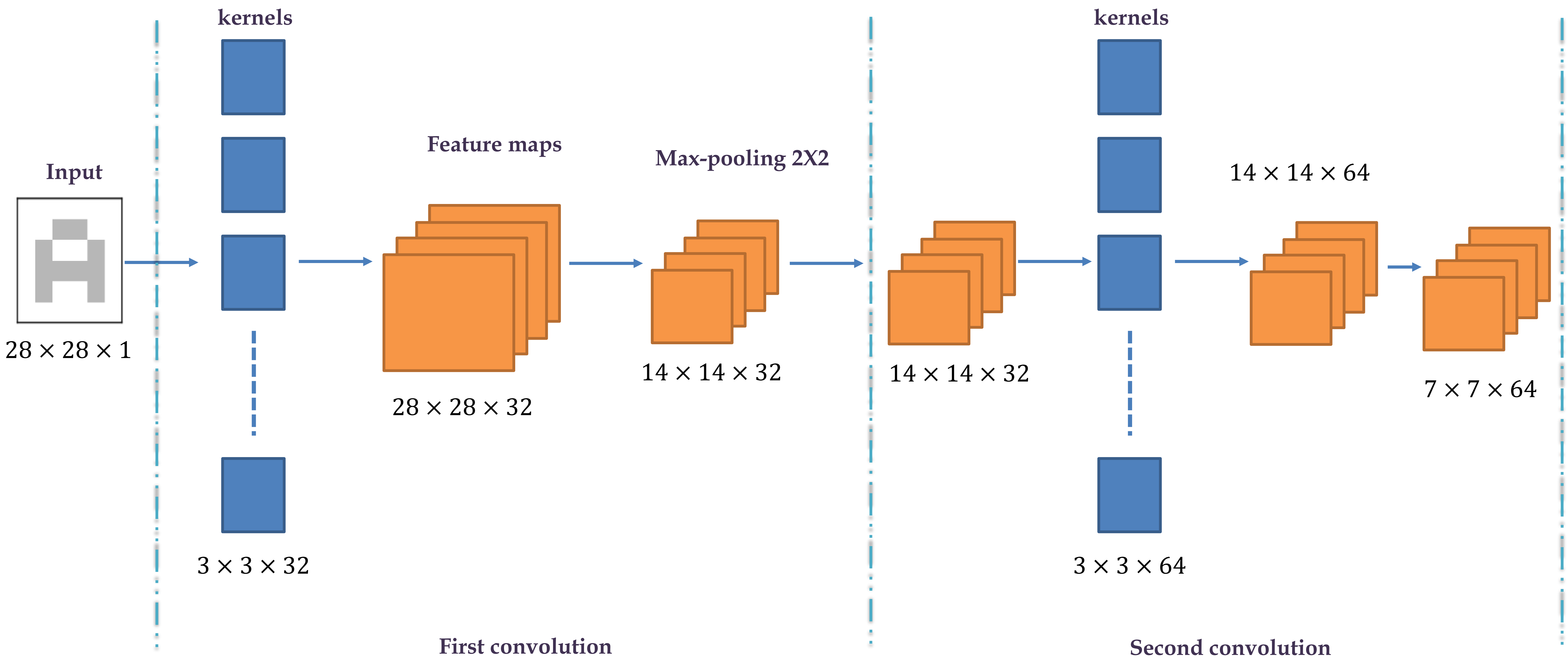


- VGG16
- ResNet50
- MobileNet
- InceptionResNetV2
- GoogleNet
- AlexNet



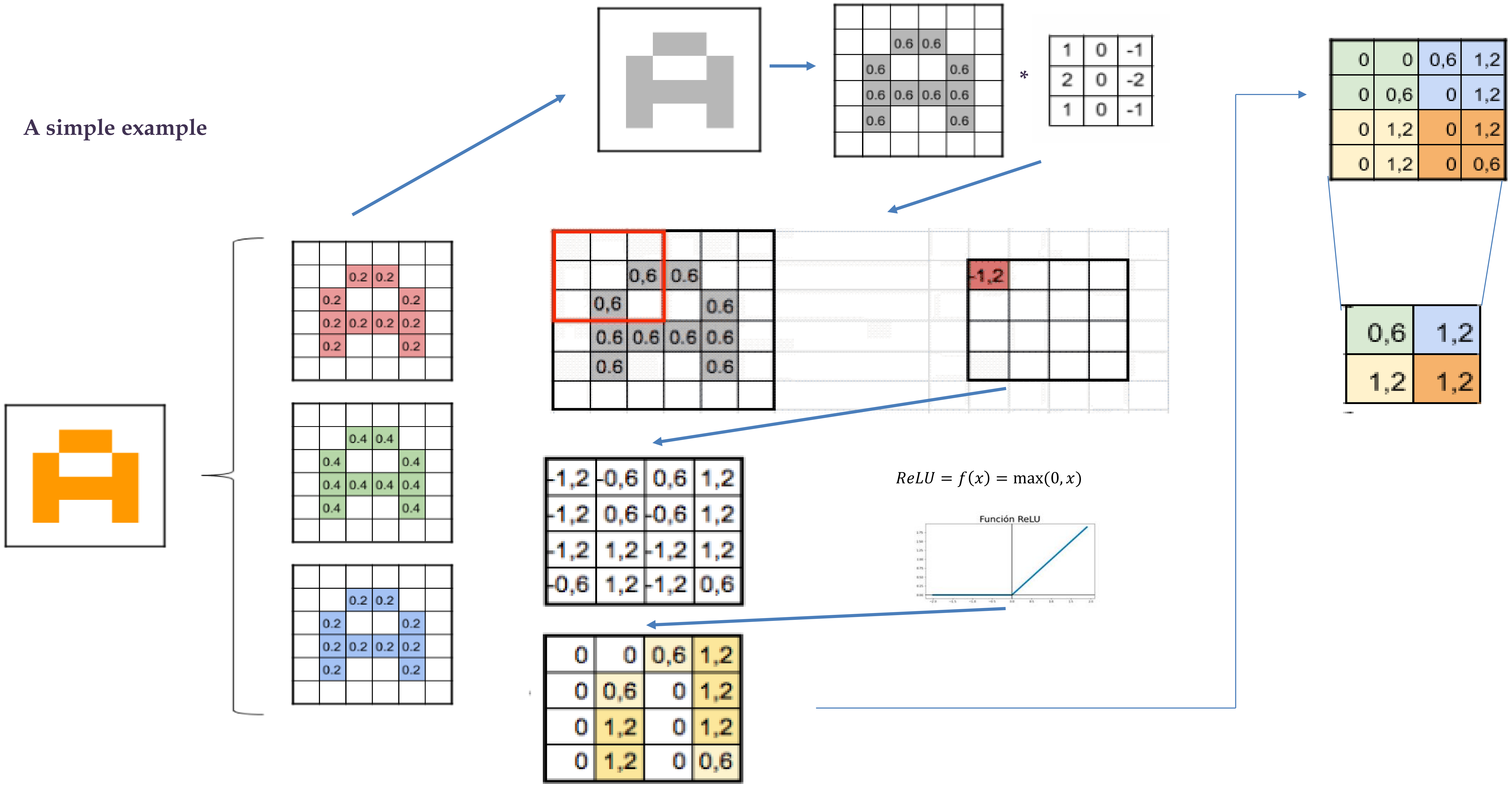
Transfer learning

Red neuronal convolucional
CNN



Red neuronal convolucional
CNN

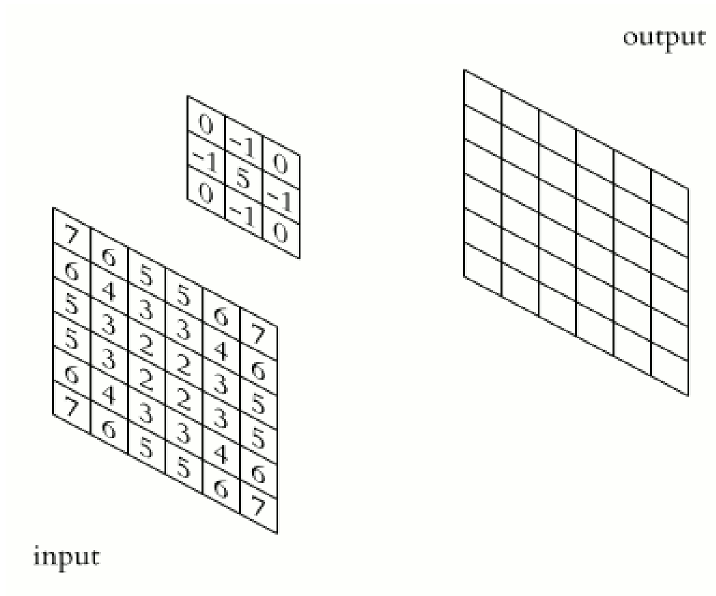
A simple example

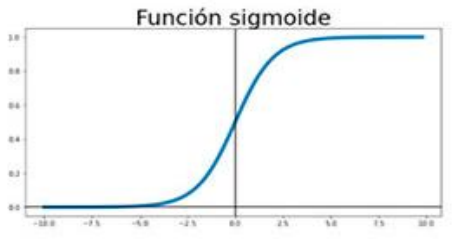


Red neuronal convolucional
CNN

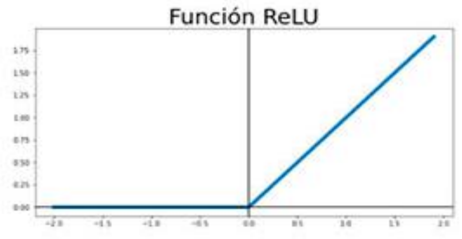
Convolutional blocks

Convolutional layers	Activation layers	Pooling layers
These layers are responsible for extracting important features from the image. They use special filters called "kernels" that slide across the image and perform mathematical operations to highlight patterns such as edges, textures, and shapes.	After convolution, an activation function is applied which introduces nonlinearity into the network. The activation function determines whether a neuron is activated or not, which helps capture complex relationships in the data.	These layers reduce the dimensionality of the features obtained by the convolutional layers. The pooling process combines the information into small regions to summarize it and retain only the most relevant information.

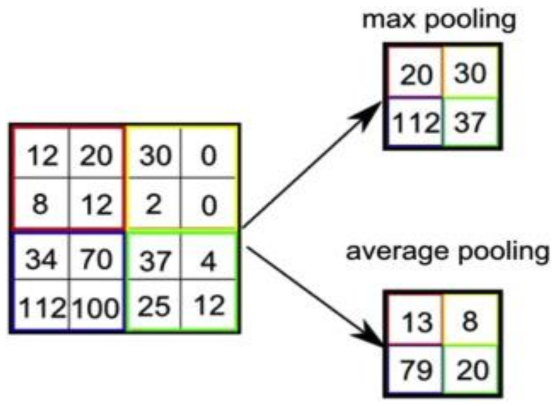




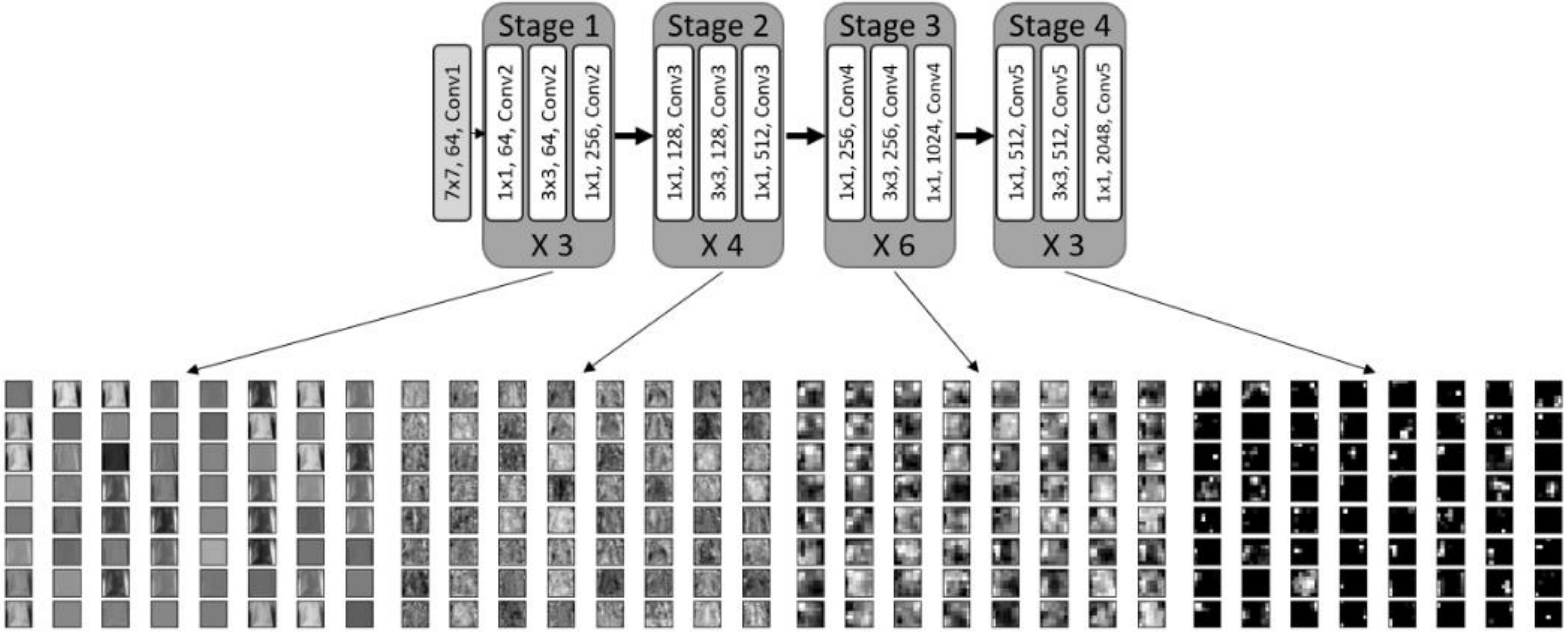
$$\sigma(x) = \frac{1}{1+e^{-x}}$$



$$f(x) = \max(0, x)$$



Red neuronal convolucional
CNN



Métodos de evaluación

Clasificadores

- Support Vector Machine (SVM)
- Logistic regression (LR)
- Nearest neighbors (KNN)
- Naïve-Bayes (NB)
- Nearest centroid (NC)

Matriz de confusión

		Predicted	
		Negative	**Positive**
Actual	**Negative**	True Negative (TN)	False Positive (FP)
	Positive	False Negative (FN)	True Positive (TP)

$$Precision = \frac{TP}{TP + FP}$$

$$Recall(sensitivity) = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$Specificity = \frac{TN}{TN + FP}$$